

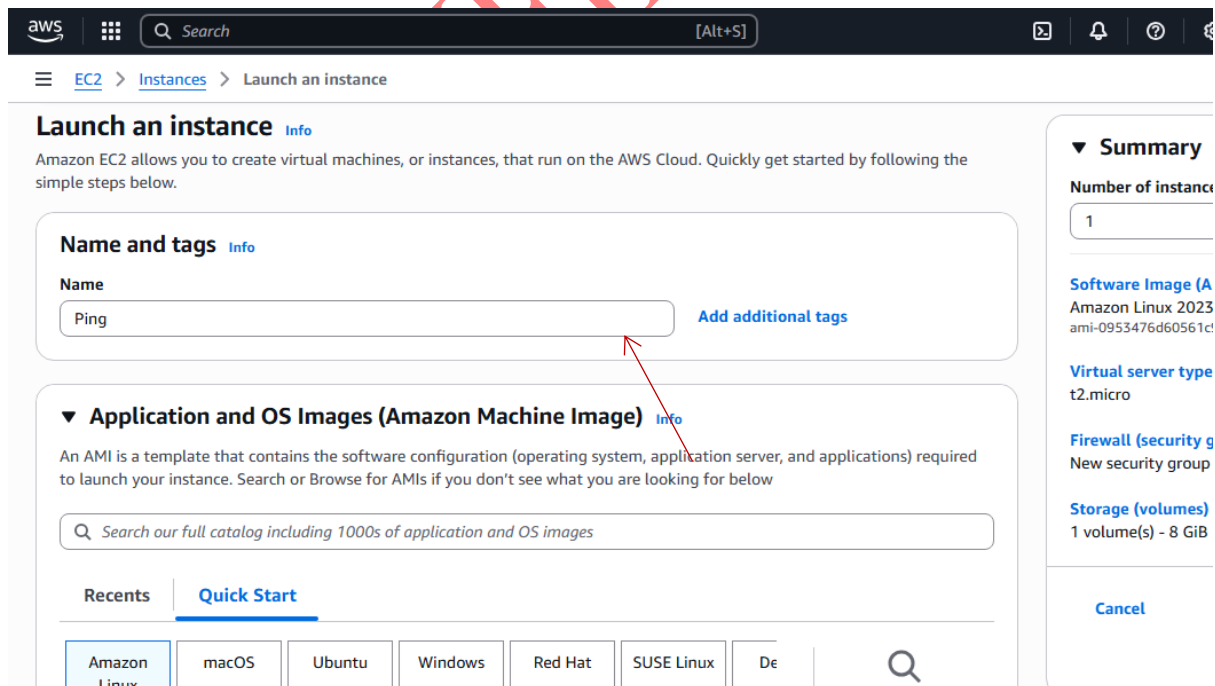
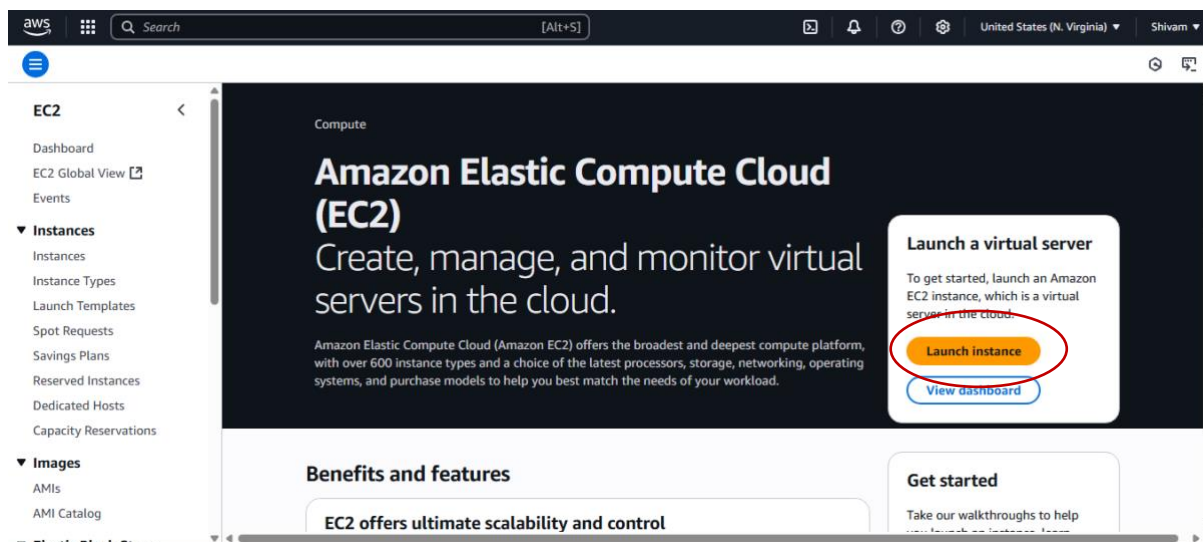
****Overview of AWS EC2 Setup****

****Environment**:**

- AWS EC2
- Region: Virginia (us-east-1)
- VPC: Default VPC
- Subnets: Subnet 1A and Subnet 1B

1. **EC2 Instances:**

- Two EC2 instances have been launched within the default VPC.
- Each instance is placed in one of the specified subnets (1A and 1B).



> Launch an instance

Create key pair

Key pair name

Key pairs allow you to connect to your instance securely.

normal

The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type

☒ RSA

RSA encrypted private and public key pair

☐ ED25519

ED25519 encrypted private and public key pair (Not supported for Windows instances)

Private key file format

☒ .pem

For use with OpenSSH

☐ .ppk

For use with PuTTY

Cancel

Create key pair

▼ Network settings Info

VPC - required Info

vpc-040ab6c8bef69ae5f
172.31.0.0/16

(default)

Subnet Info

subnet-032be374230b598e4
VPC: vpc-040ab6c8bef69ae5f Owner: 746491137895
Availability Zone: us-east-1a Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.0.0/20

Create new subnet

Auto-assign public IP Info

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) Info

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ Create security group

☐ Select existing security group

Security group name - required

launch-wizard-3

This security group will be added to all network interfaces. The name can't be edited after the security group is created. Max length is 255

Inbound Security Group Rules

▼ Security group rule 1 (TCP, 3389, 0.0.0.0/0)

Type: **rdp** | Protocol: **TCP** | Port range: **3389**

Source type: **Anywhere** | Source: **0.0.0.0/0** | Description - optional: **e.g. SSH for admin desktop**

Rules with source of 0.0.0.0/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Summary

Number of instances: 1

Software Image (AMI): Microsoft Windows Server 2016 ...read more

Virtual server type (instance type): t2.micro

Firewall (security group): New security group

Storage (volumes): 1 volume(s) - 30 GiB

Launch instance

2. **Network Configuration**:

- Both instances are configured to communicate within the same VPC.
- Security groups are set to allow ICMP traffic for pinging.

EC2 > Instances

Instances (1/1)

Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
Ping	i-0f5af918085649142	Running	t2.micro	Initializing	View alarms +	us-east-1a

i-0f5af918085649142 (Ping)

Security

Instance summary

Instance ID: i-0f5af918085649142

Public IPv4 address: 18.209.227.137 | open address

Private IPv4 addresses: 172.31.11.215

- **Security Groups**:

- Ensure that the security group associated with each instance allows inbound ICMP traffic (for ping).

EC2 > Security Groups > sg-06bcfa3296cc4273a - launch-wizard-3 > Edit inbound rules

Edit inbound rules

Inbound rules control the incoming traffic that's allowed to reach the instance.

Inbound rules

Security group rule ID	Type	Protocol	Port range	Source	Description - optional
sg-03700497ac5f98f13	RDP	TCP	3389	Cus...	
-	All ICMP - IPv4	ICMP	All	An...	

Rules with source of 0.0.0.0/0 or ::/0 allow all IP addresses to access your instance. We recommend setting security group rules to allow access from known IP addresses only.

Save rules

EC2 > Instances > Launch an instance

Name and tags [Info](#)

Name: [Add additional tags](#)

Application and OS Images (Amazon Machine Image) [Info](#)

An AMI is a template that contains the software configuration (operating system, application server, and applications) required to launch your instance. Search or browse for AMIs if you don't see what you are looking for below

Recents **Quick Start**

Amazon Linux macOS Ubuntu **Windows** Red Hat SUSE Linux De

[Browse more AMIs](#)
Including AMIs from AWS, Marketplace and the Community

Summary

Number of instances: [Info](#)

Software Image (AMI)
Amazon Linux 2023 AMI 2023.7.2...[read more](#)
ami-095347bd60561c955

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 8 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

- **Subnet Configuration**:

- Verify that both subnets (1A and 1B) are correctly configured and associated with the same route table.

EC2 > Instances > Launch an instance

Key pair name - required

Normal [Create new key pair](#)

For Windows instances, you use a key pair to decrypt the administrator password. You then use the decrypted password to connect to your instance.

Network settings [Info](#)

VPC - required [Info](#)

vpc-040ab6c8bef69ae5f (default) [Create new VPC](#)

Subnet [Info](#)

subnet-0b53412ab143411c2 [Create new subnet](#)

VPC: vpc-040ab6c8bef69ae5f Owner: 746491137895
Availability Zone: us-east-1b Zone type: Availability Zone
IP addresses available: 4091 CIDR: 172.31.80.0/20

Auto-assign public IP [Info](#)

Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)

A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

Summary

Number of instances: [Info](#)

Software Image (AMI)
Microsoft Windows Server 2016 ...[read more](#)
ami-08ded310ca86fa861

Virtual server type (instance type)
t2.micro

Firewall (security group)
New security group

Storage (volumes)
1 volume(s) - 30 GiB

[Cancel](#) [Launch instance](#) [Preview code](#)

EC2 > Instances

Instances (2) [Info](#)

Last updated less than a minute ago [Connect](#) [Instance state](#) [Actions](#) [Launch instances](#)

[All states](#)

☐	Name	Instance ID	Instance state	Instance type	Status check	Alarm status	Availability
☐	Ping2	i-003381dbfc414d855	Running	t2.micro	Initializing	View alarms +	us-east-1b
☐	Ping	i-0f5af918085649142	Running	t2.micro	2/2 checks passed	View alarms +	us-east-1a

Select an instance

EC2 > Instances > i-Of5af918085649142 > Connect to instance

Connect Info

Connect to an instance using the browser-based client.

Session Manager **RDP client** EC2 serial console

Record RDP connections

You can now record RDP connections using AWS Systems Manager just-in-time node access. [Learn more](#)

Try for free

Instance ID

i-Of5af918085649142 (Ping)

Connection Type

Connect using RDP client

Download a file to use with your RDP client and retrieve your password.

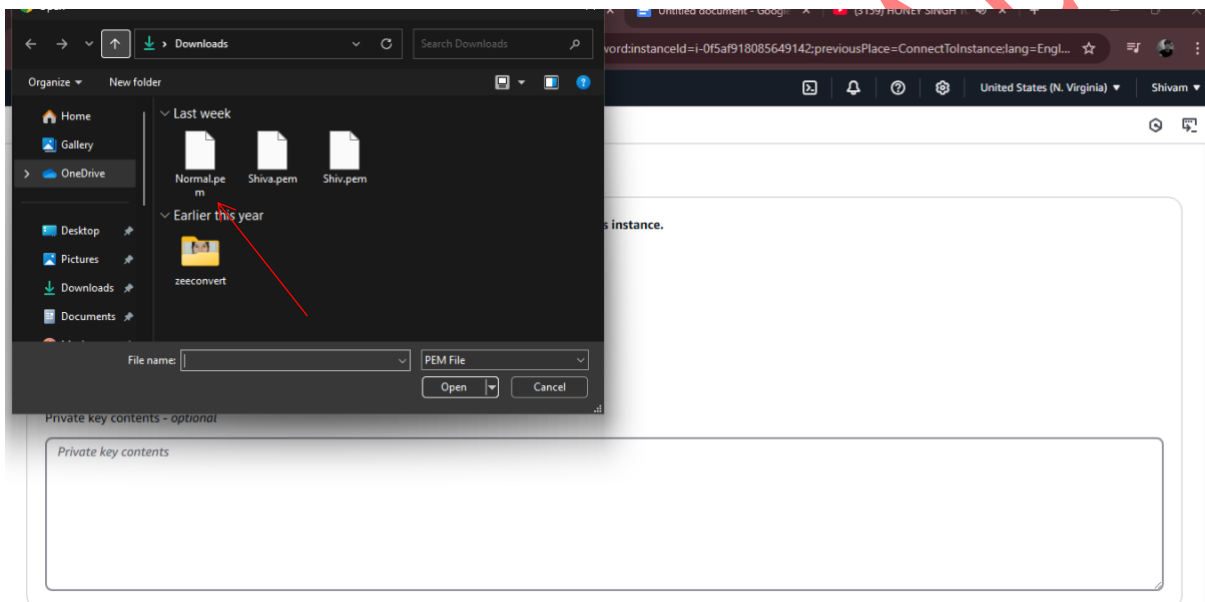
Connect using Fleet Manager

To connect to the instance using Fleet Manager Remote Desktop, the SSM Agent must be running on the instance. For more information, see [Working with SSM Agent](#)

You can connect to your Windows instance using a remote desktop client of your choice, and by downloading and running the RDP shortcut file below:

[Download remote desktop file](#)

When prompted, connect to your instance using the following username and password:



EC2 > Instances > i-Of5af918085649142 > Get Windows password

Instance ID

i-Of5af918085649142 (Ping)

Key pair associated with this instance

Normal

Private key

Either upload your private key file or copy and paste its contents into the field below.

[Upload private key file](#)

Normal.pem
1.674KB

Private key contents - optional

```
-----BEGIN RSA PRIVATE KEY-----
MIIEowIBAAKCAQEAvgH5/Oz0FqZafD94WE9obfVJ10fANyyEoYL5j3B/XGy+IrO
xEyuWQaBtagDUhBcl+ODzhePsmK2tyYfKirYUpep6XCutuhQcT4clMh11ZXxrJx
35kr+YDRIF9NZC4JknRoPjLvlRM2DNK8c2vq38barYnoB2FFTC+ISmRm798z+HB5
Wm4Klsr/DPdIB/PAer62Mwcvx3LO2b6mivHeJKb8P3Ya1W8CUuo/aJeCP3ZJd8N
Aldyqj+XH50nGwXmFOLTVObvwwC1857uKWIQ/SUapFDk5a8XWttUF4slBXt289c
EWPhRuWHKwK7Wk2ToDYw71BpD1AqU30wQ/bpQIDAQABADgW5x2dKAof0u
GONy7zhMFPiorihwFX4oqlgRprR71AFteanNSpgENolyBZIUQpTkt0uAZ07bXf
-----
```

Cancel

Decrypt password

The first screenshot shows the AWS console 'Connect to instance' page for instance `i-0f5af918085649142`. It provides instructions on how to connect using an RDP client and shows the public DNS `ec2-18-209-227-137.compute-1.amazonaws.com`. A password is displayed as `fKQxCTEtqM4?fc%(pL5@!UVmIHwp&Dor`. A red arrow points from the password field in the console to the terminal window in the second screenshot.

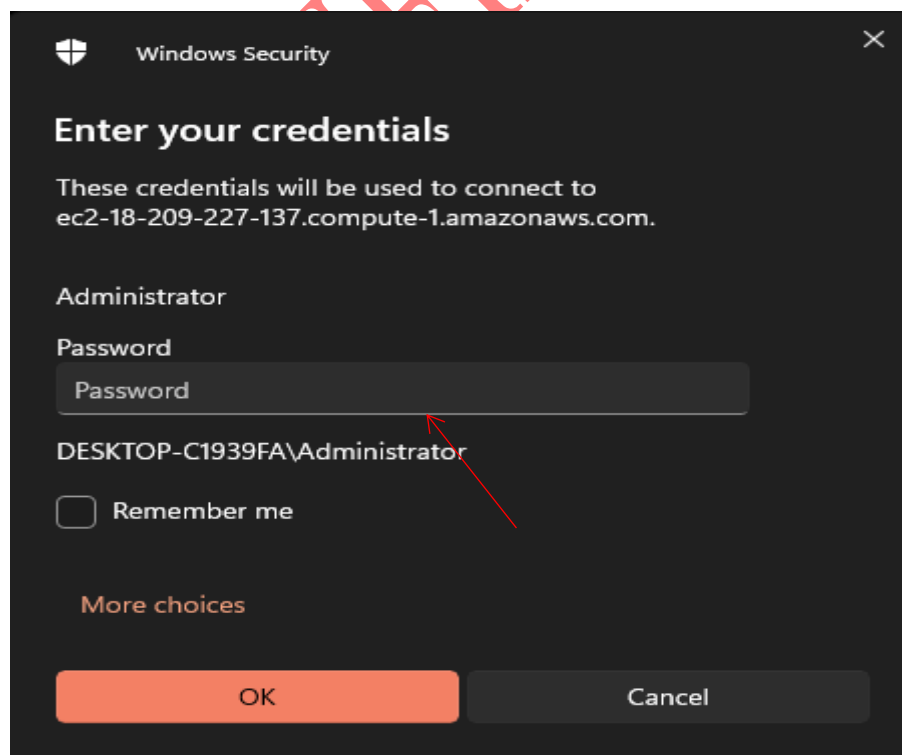
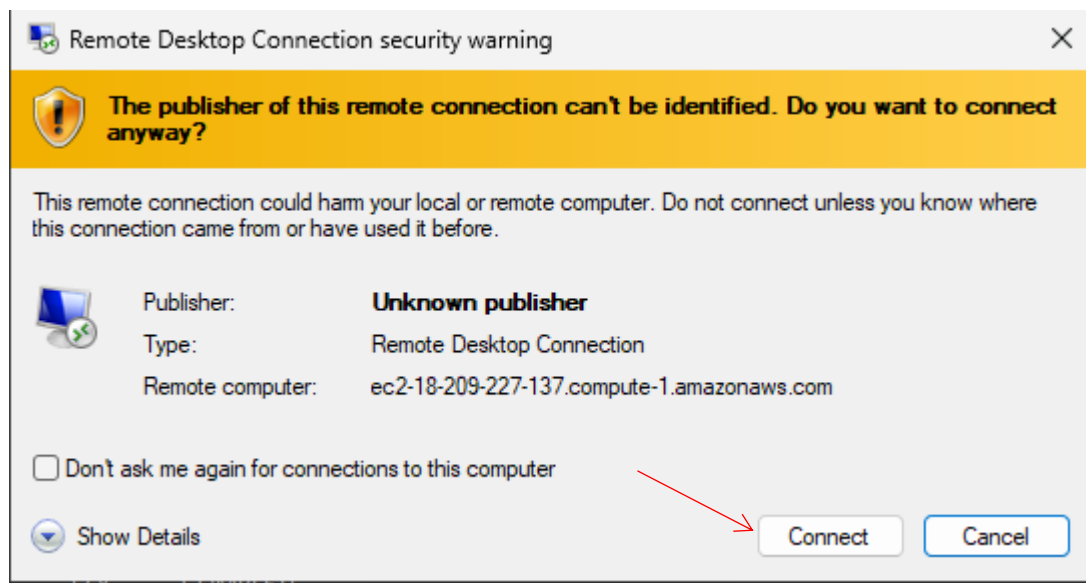
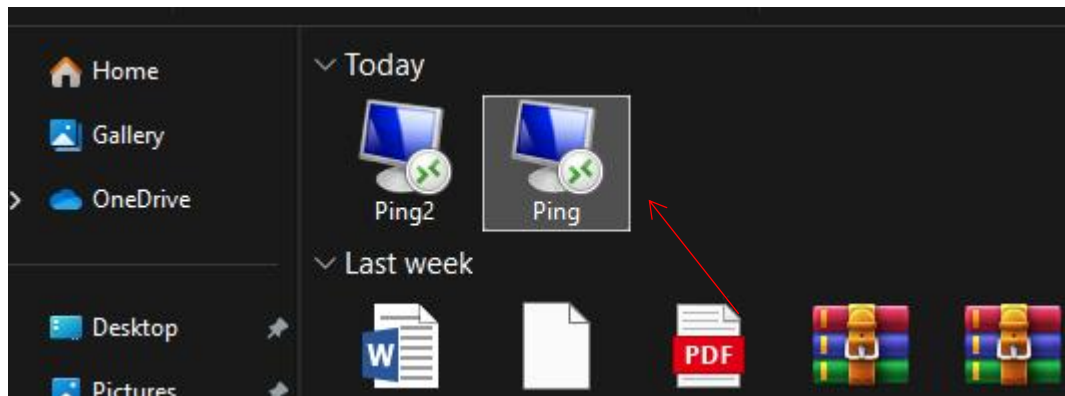
The second screenshot shows the AWS console 'Instances' page. A table lists two instances: 'Ping2' (ID: `i-003381dbfc414d855`) and 'Ping' (ID: `i-0f5af918085649142`). Both are in a 'Running' state. A red arrow points from the 'Connect' button for the 'Ping' instance to the terminal window in the third screenshot.

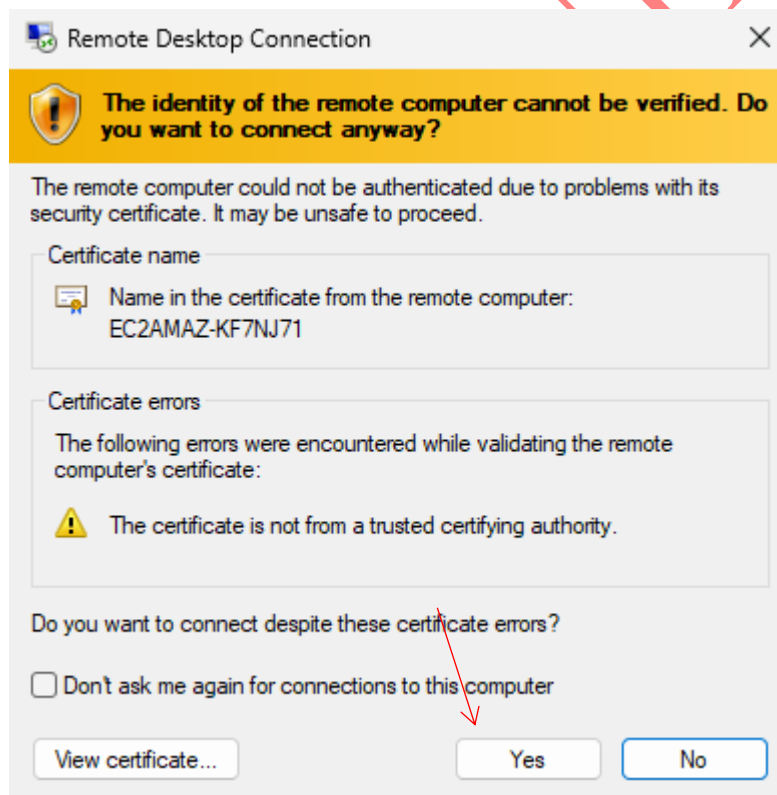
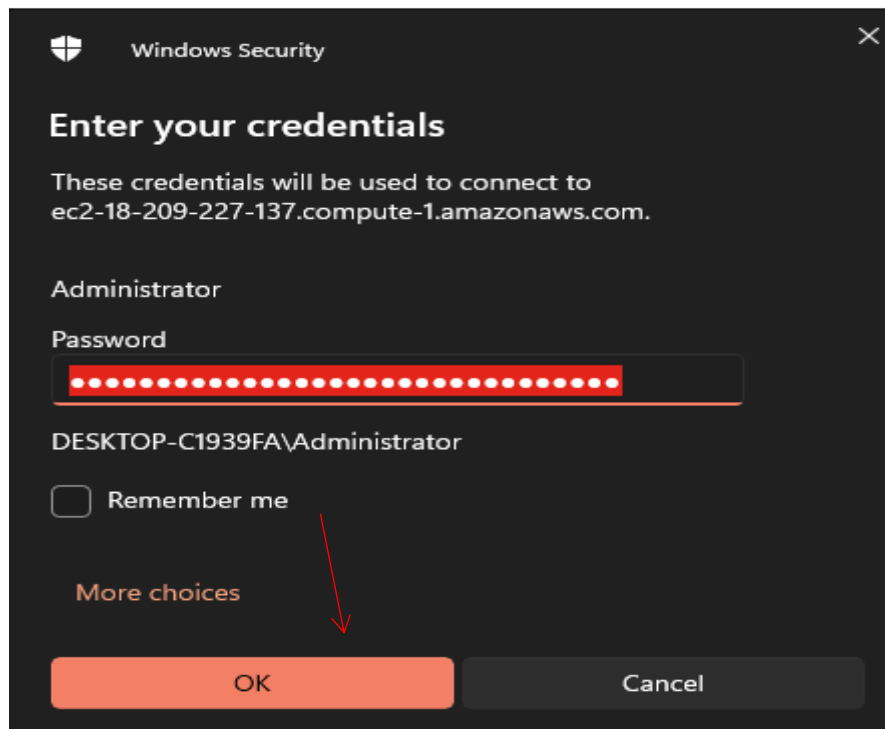
The third screenshot shows the 'Connect to instance' page for instance `i-003381dbfc414d855`. It shows the public DNS `ec2-54-163-196-236.compute-1.amazonaws.com` and a password `fKQxCTEtqM4?fc%(pL5@!UVmIHwp&Dor`. A red arrow points from the password field in the console to the terminal window in the fourth screenshot.

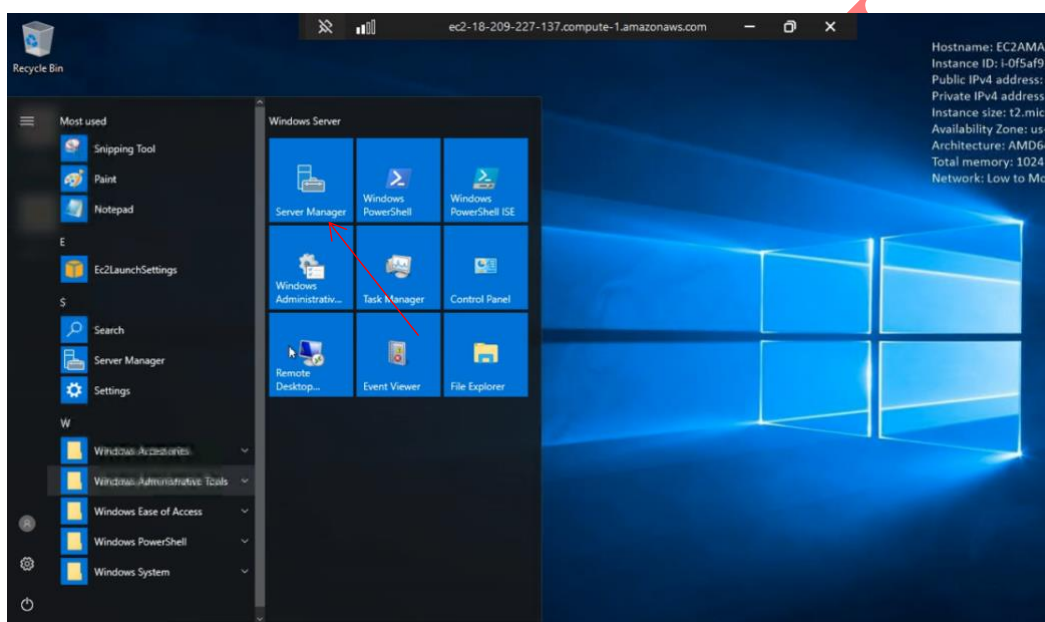
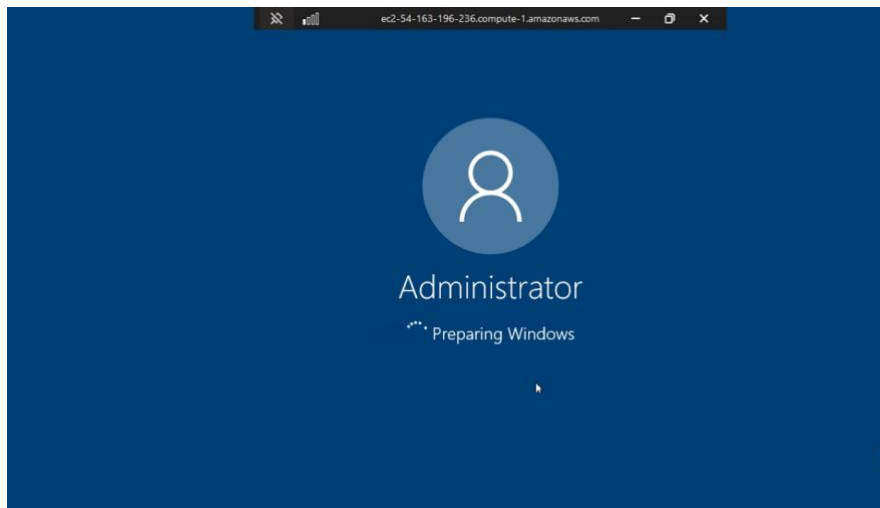
The fourth screenshot shows a terminal window with the password `fKQxCTEtqM4?fc%(pL5@!UVmIHwp&Dor` entered. Below it, the output of a ping command is shown: `ping` and `ping2`. A red arrow points from the password field in the console to the terminal window.

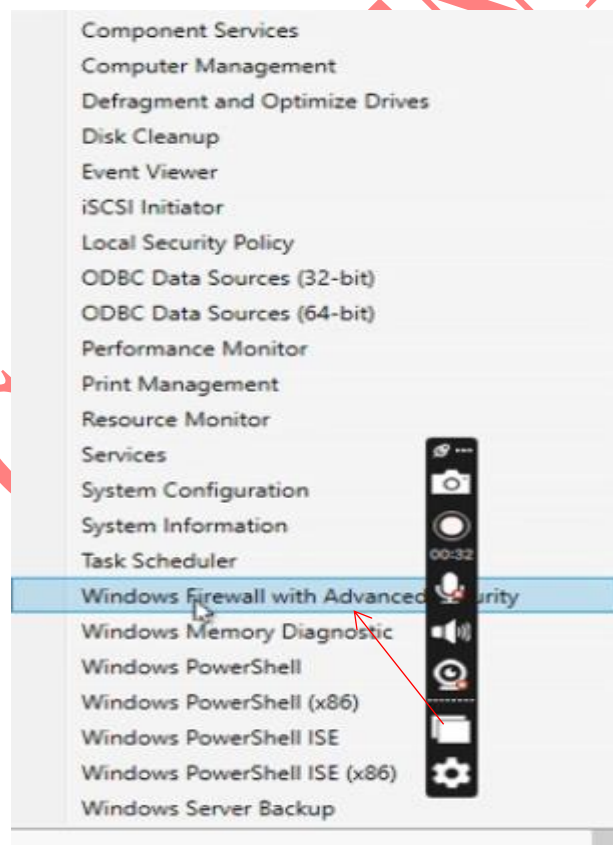
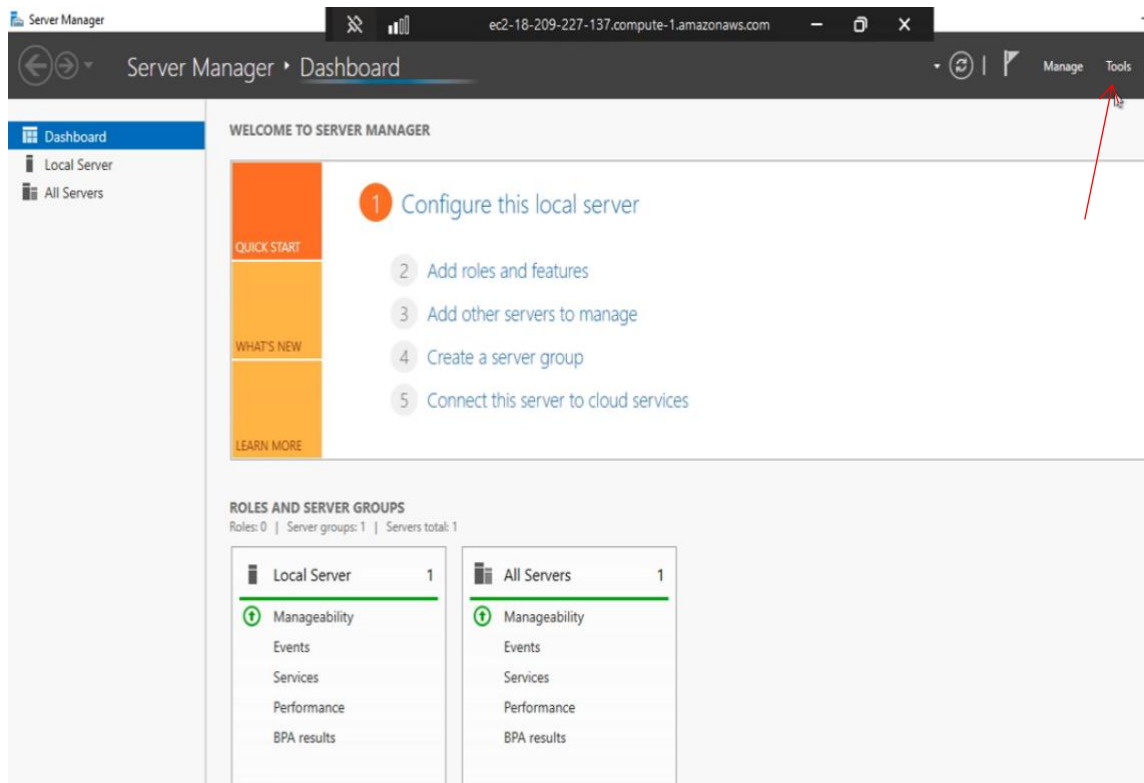
3. **Ping Test**:

- A ping command was executed from one EC2 instance to the other to verify connectivity.
- Successful replies indicate that the instances can communicate over the network.



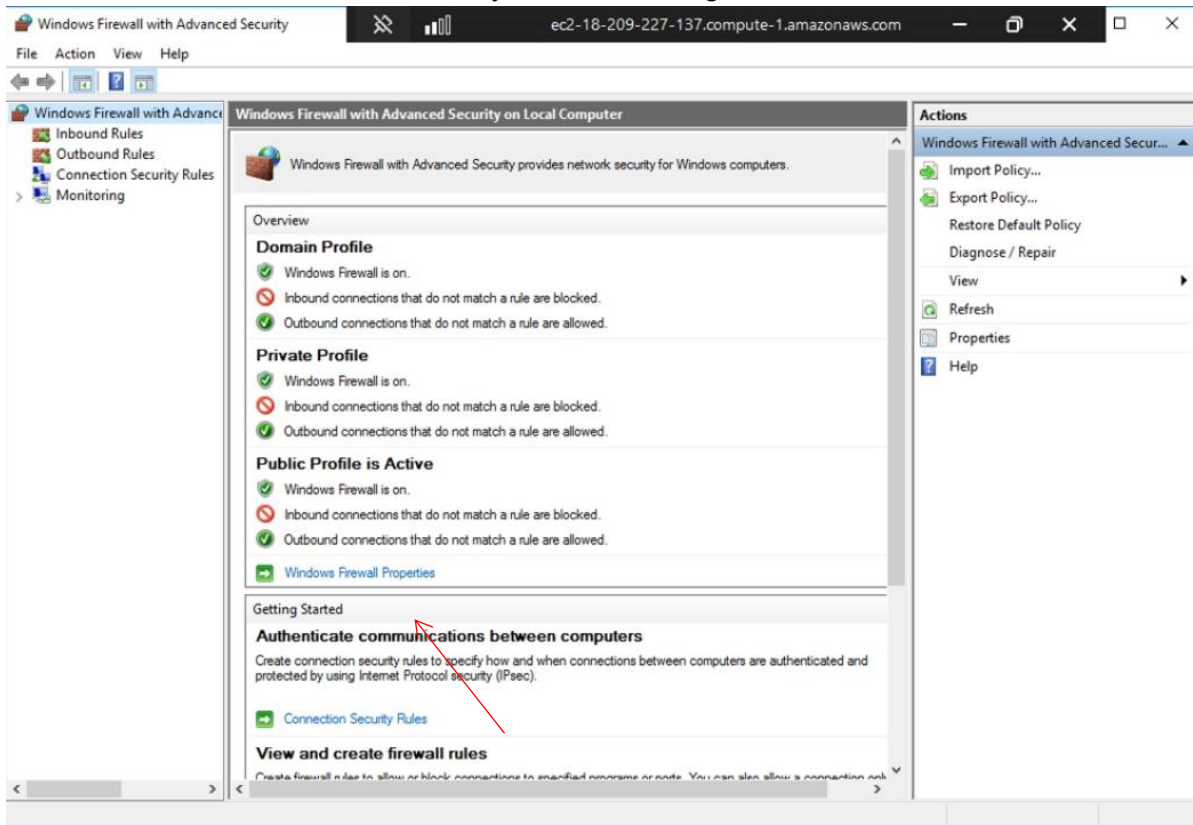


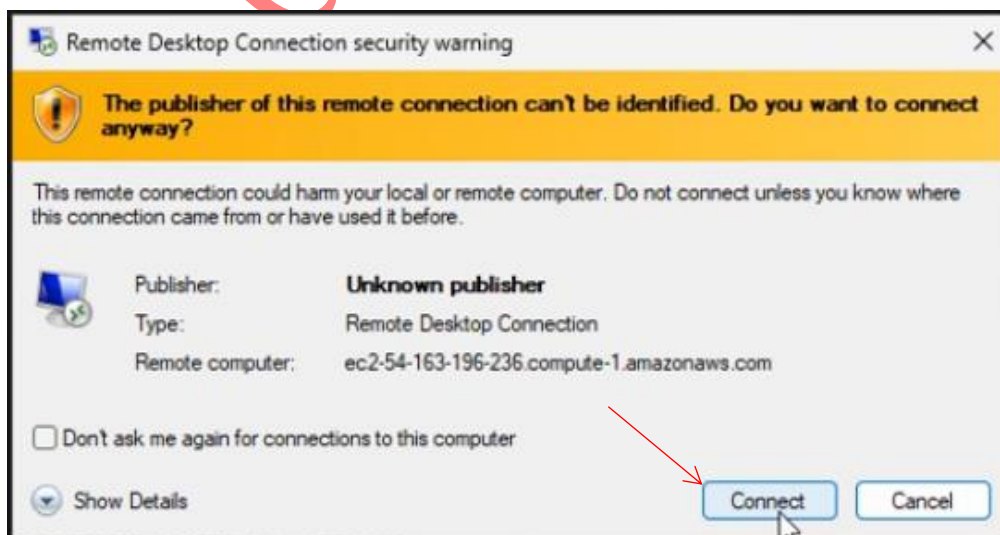
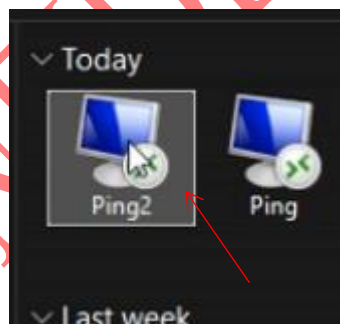
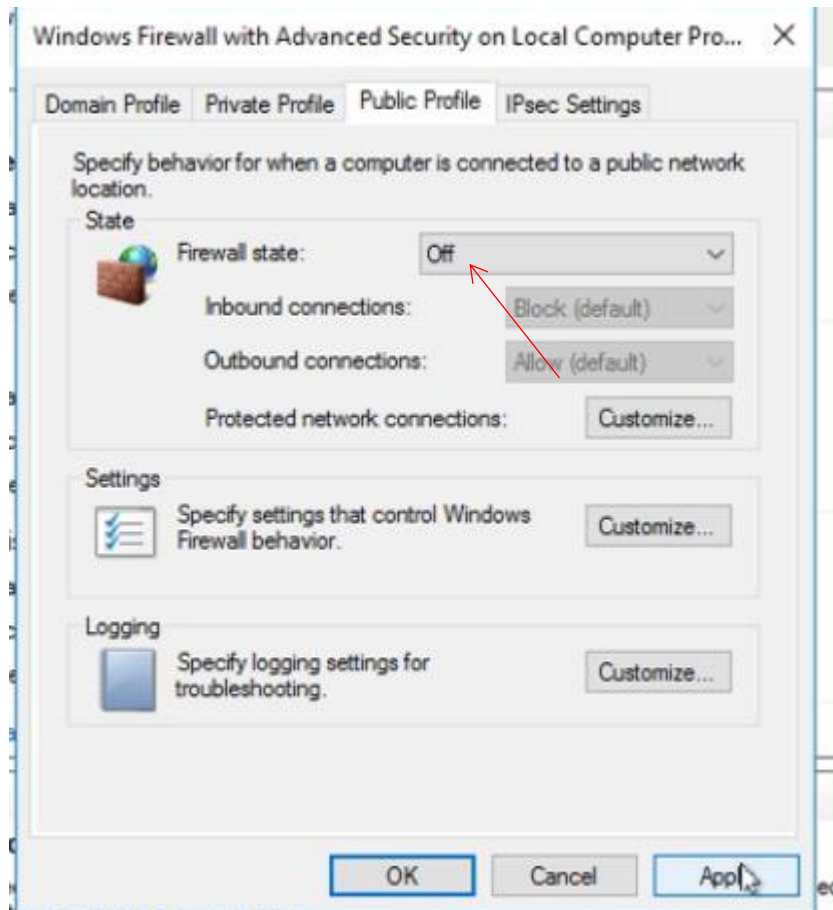


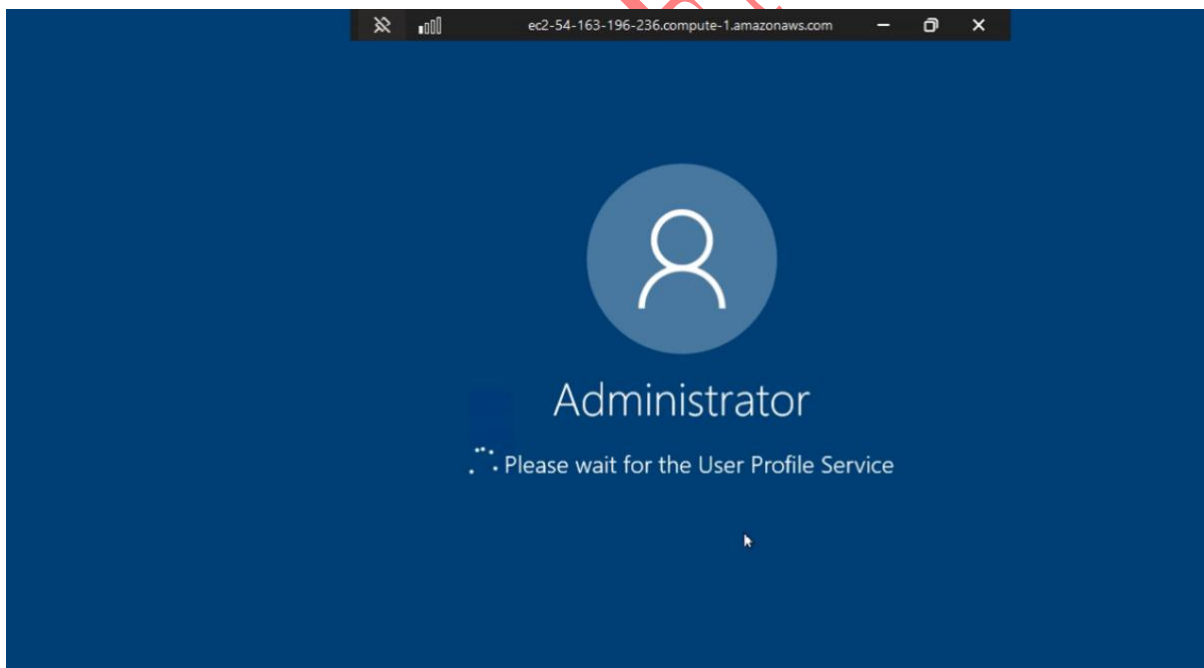
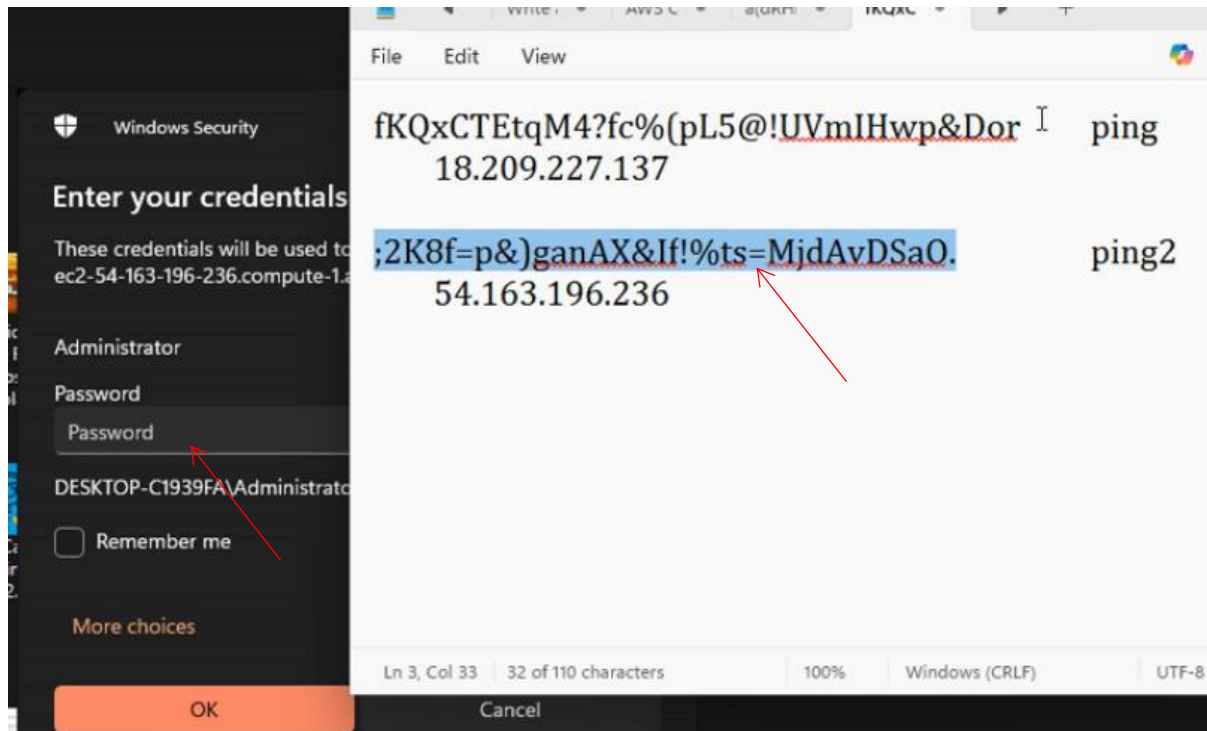


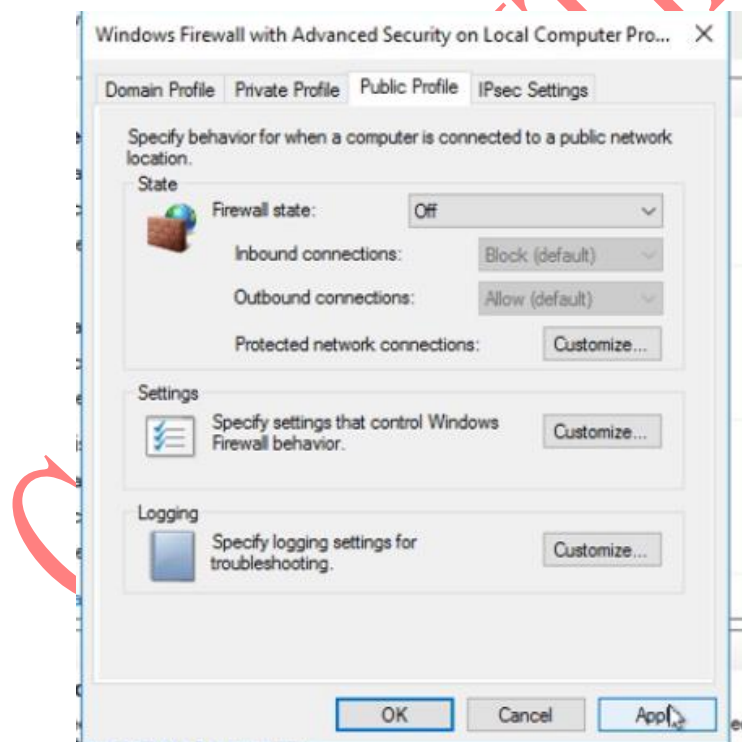
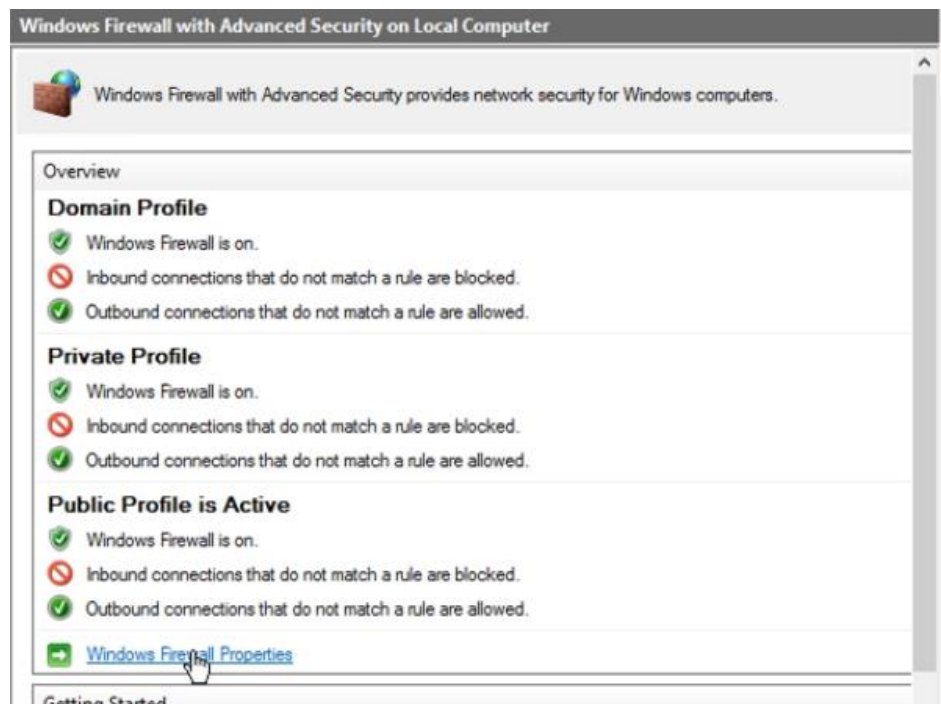
‘- ****Network ACLs****:

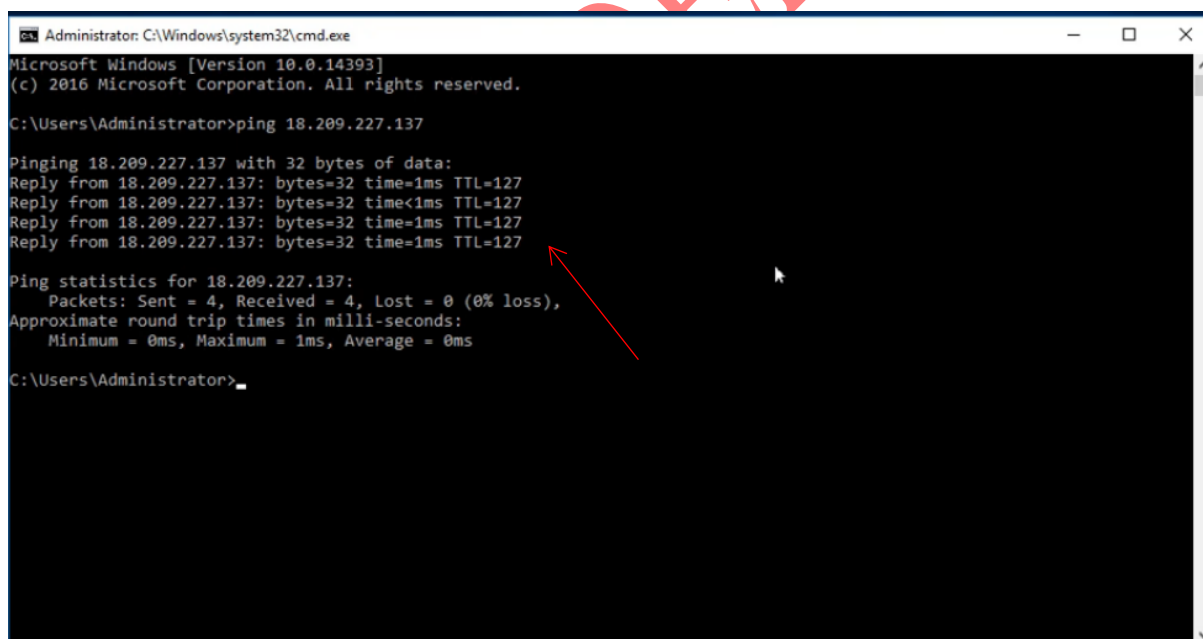
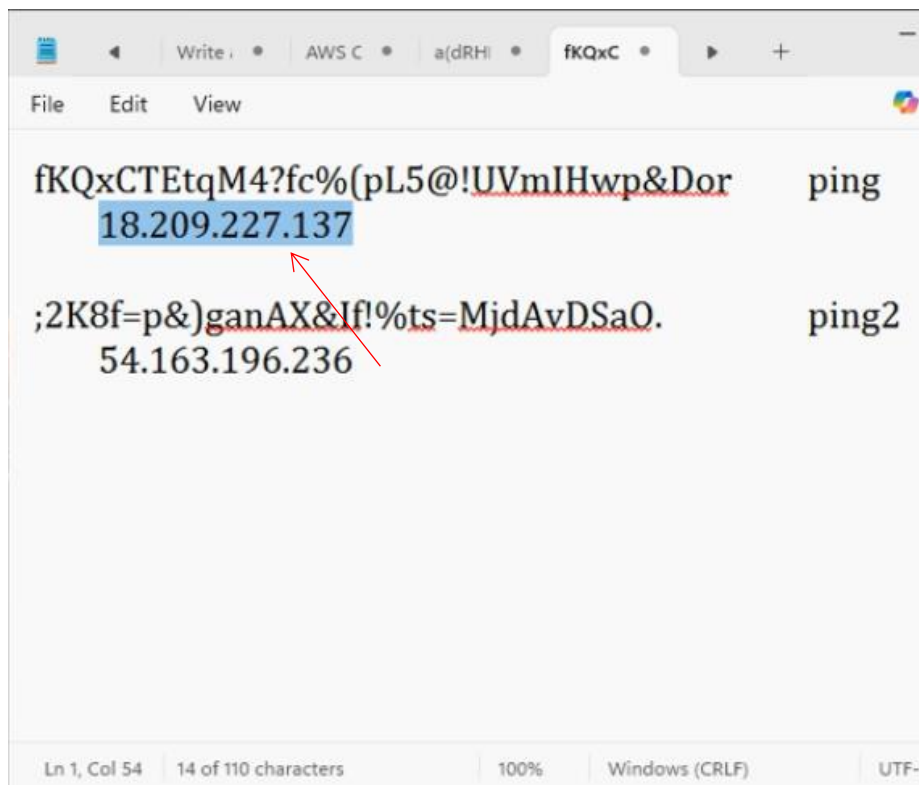
- Check Network ACLs to ensure they are not blocking ICMP traffic.











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